Delay Formula

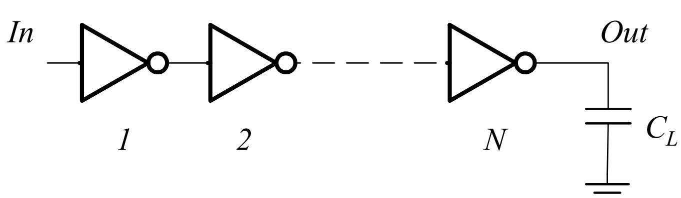
$$Delay \sim R_W \left(C_{int} + C_L\right)$$

$$\boxed{t_p = kR_W C_{\text{in}} \left(C_{\text{int}} / C_{in} + C_L / C_{in}\right) = t_{inv} \left(\gamma + f\right)}$$

$C_{int} = \gamma\, C_{in}$ ($\gamma \approx 1$ for CMOS inverter)
$f = C_L/C_{in}$ – electrical fanout
$t_{\text{inv}} = kR_{min}C_{ginv}$

t_{inv} is independent of sizing of the gate!!!

Apply to Inverter Chain



$$t_p = t_{p1} + t_{p2} + \ldots + t_{pN}$$

$$t_{pj} = t_{inv}\left(\gamma + \frac{C_{in,j+1}}{C_{in,j}}\right)$$

$$t_p = \sum_{j=1}^{N} t_{p,j} = t_{inv} \sum_{i=1}^{N} \left(\gamma + \frac{C_{in,j+1}}{C_{in,j}}\right), \quad C_{in,N+1} = C_L$$

Optimal Tapering for Given N

- Delay equation has N-1 unknowns, $C_{in,2} \ldots C_{in,N}$
- **To minimize the delay,** find N-1 partial derivatives:

$$t_p = \ldots + t_{inv}\frac{C_{in,j}}{C_{in,j-1}} + t_{inv}\frac{C_{in,j+1}}{C_{in,j}} + \ldots$$

$$\frac{dt_p}{dC_{in,j}} = t_{inv}\frac{1}{C_{in,j-1}} - t_{inv}\frac{C_{in,j+1}}{C_{in,j}^2} = 0$$

Optimal Tapering for Given N (cont'd)

- Result: every stage has **equal fanout**:

$$\boxed{\frac{C_{in,j}}{C_{in,j-1}} = \frac{C_{in,j+1}}{C_{in,j}}}$$

- In other words, size of each stage is geometric mean of two neighbors:

$$\boxed{C_{in,j} = \sqrt{C_{in,j-1}C_{in,j+1}}}$$

- Equal fanout → every stage will have same delay

Optimum Delay and Number of Stages

- When each stage has same fanout f:

$$f^N = F = C_L / C_{in,1}$$

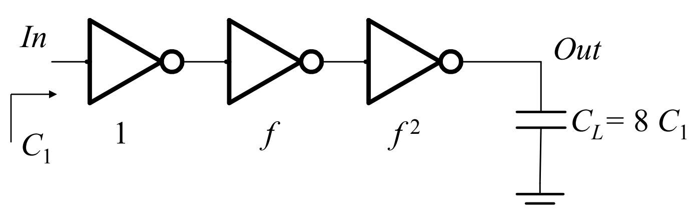
- Effective fanout of each stage:

$$f = \sqrt[N]{F}$$

- Minimum path delay:

$$t_p = N t_{inv} \left(\gamma + \sqrt[N]{F} \right)$$

Example



C_L/C_1 has to be evenly distributed across $N = 3$ stages:

$$f = \sqrt[3]{8} = 2$$

Delay Optimization Problem #2

- You are given:
  - The size of the first inverter
  - The size of the load that needs to be driven
- Your goal:
  - Minimize delay by finding optimal number and sizes of gates
- So, need to find N that minimizes:

$$t_p = N t_{inv}\left(\gamma + \sqrt[N]{C_L / C_{in}}\right)$$

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Solving the Optimization

- Rewrite N in terms of fanout/stage f:

$$f^N = C_L / C_{in} \rightarrow \boxed{N = \frac{\ln\left(C_L / C_{in}\right)}{\ln f}}$$

$$t_p = N t_{inv}\left(\left(C_L / C_{in}\right)^{1/N} + \gamma\right) = t_{inv} \ln\left(C_L / C_{in}\right)\left(\frac{f+\gamma}{\ln f}\right)$$

$$\frac{\partial t_p}{\partial f} = t_{inv} \ln\left(C_L / C_{in}\right) \cdot \frac{\ln f - 1 - \gamma / f}{\ln^2 f} = 0$$

$$f = \exp\left(1 + \gamma / f\right)$$

$\boxed{\text{For } \gamma = 0, f = e, N = \ln (C_L/C_{in})}$

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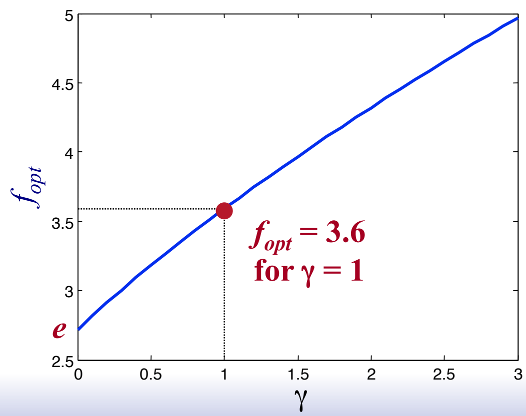
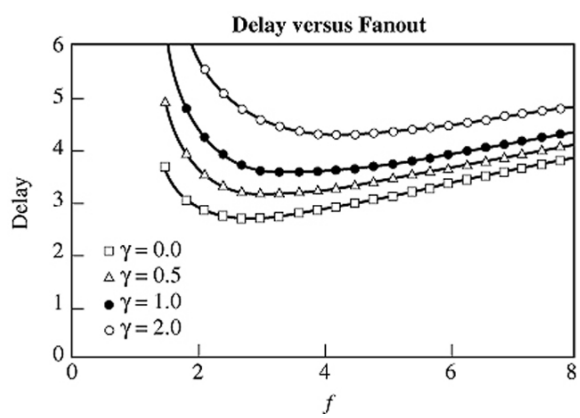
Optimum Effective Fanout f

- Optimum f for given process defined by γ

$$f = \exp(1 + \gamma / f)$$



In Practice: Plot of Total Delay



[Hodges, p.281]

- Curves very flat for $f > 2$
  - Simplest/most common choice: $f = 4$

Normalized Delay As a Function of F

$$t_p = N t_{inv}\left(\gamma + \sqrt[N]{F}\right), \; F = C_L / C_{in}$$

F	Unbuffered	Two Stage	Inverter Chain
10	11	8.3	8.3
100	101	22	16.5
1000	1001	65	24.8
10,000	10,001	202	33.1

(γ = 1)

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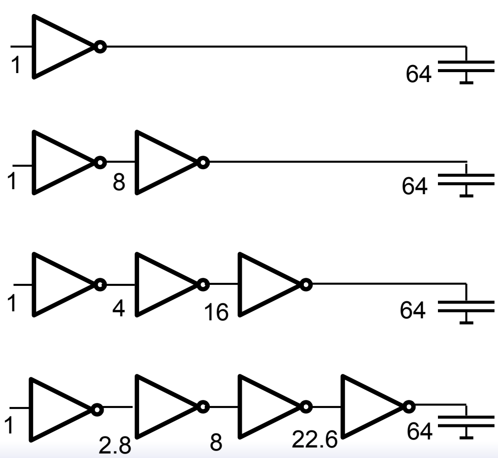
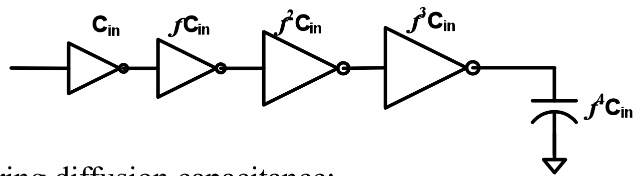
Buffer Design



N	*f*	t_p
1	64	65
2	8	18
3	4	15
4	2.8	15.3

What About Energy (and Area)?



Ignoring diffusion capacitance:

$$C_{tot} = C_{in} + f \cdot C_{in} + \ldots + f^N \cdot C_{in}$$
$$= C_{in} \cdot (1 + f + \ldots + f^N)$$
$$= C_{in} + C_{in} \cdot f^N + C_{in} \cdot \underbrace{f \cdot (1 + f + \ldots + f^{N-2})}_{\text{Overhead !!! } f(f^{N-1}-1)/(f-1)}$$

Example (γ=0): C_L = 20pF; C_i = 50fF → N = 6
Fixed: 20pF
Overhead: 11.66pF !!!

Example Overhead Numbers

Example: C_L = 20pF; C_{in} = 50fF